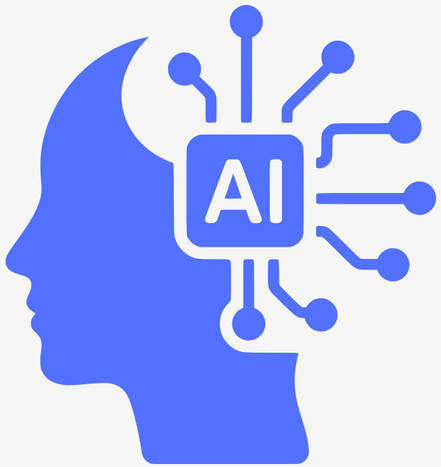
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TEXT-TO-SQL ARCHITECTURES

IMAGINE ASKING YOUR DATABASE A QUESTION IN SIMPLE ENGLISH, AND AI CONVERTS IT INTO SQL, FETCHES THE CORRECT DATA, AND GIVES YOU THE ANSWER. BUT WHAT IF AI GENERATES A DANGEROUS QUERY?

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WHAT IS TEXT-TO-SQL?

- Text-to-SQL converts natural language questions into SQL queries.
- Users can ask questions in simple English instead of writing SQL manually.
- AI understands the question, generates SQL, and retrieves the required data.

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SAFE QUERYING

- AI-generated SQL can be incorrect or dangerous.
- Allow only approved operations, preferably read-only queries.
- Block dangerous commands such as DROP, DELETE, and unauthorized UPDATE.
- Use database permissions, query validation, and execution limits.
- Never expose database credentials or sensitive data to the user.

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SCHEMA UNDERSTANDING

- A database schema describes tables, columns, data types, and relationships.
- AI needs to understand the schema before generating SQL.
- Provide relevant table structures and relationships to the AI.
- Better schema understanding leads to more accurate queries.

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SQL AGENTS

- A SQL agent uses an LLM to answer questions by interacting with a database.
- It can understand a question, inspect the schema, generate SQL, and execute an approved query.
- It can review errors and correct a query when appropriate.
- Every query must pass security checks before execution.

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REAL-WORLD APPLICATIONS

- Business analytics
- Sales and revenue reports
- Customer insights
- Inventory management
- Financial reporting
- Natural language database assistants